

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Engineering (ME784)

INDEX

Sr. No.	Date	Title	No. of Pages	Marks	Date of Assessment	Sign of Faculty
1		Introduction to CAE				
2		Study and Application of Computer Graphics in CAD				
3		Algorithms Used for Generating 2D Output Primitives				
4		Geometric Transformations				
5		CAE of Manufacturing Processes				
6		Discretization Methods				
7		Numerical Solution Methods for System of Linear Equations				
8		Numerical programming of Heat Transfer Phenomena				
9		Reverse Engineering				
10		Rapid Prototyping				

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Engineering (ME784)

Date:

EXPERIMENT NO. 1
INTRODUCTION TO COMPUTER AIDED ENGINEERING

AIM:

Introduction to Computer Aided Engineering (CAE).

OBJECTIVES:

1. Introduction to CAE and its role in the industry.
2. To know the necessity of CAE software's in product life cycle.

THEORY:

Computer-aided engineering (CAE) is the broad usage of computer software to aid in engineering tasks. It includes computer-aided design (CAD), computer-aided analysis (CAA), computer-integrated manufacturing (CIM), computer-aided manufacturing (CAM), material requirements planning (MRP), and computer-aided planning (CAP).

CAE applications support a wide range of engineering disciplines or phenomena including:

- Stress and dynamics analysis on components and assemblies using finite element analysis (FEA)
- Thermal and fluid analysis using computational fluid dynamics (CFD)
- Kinematics and dynamic analysis of mechanisms (multibody dynamics)
- Mechanical event simulation (MES)
- Control systems analysis
- Simulation of manufacturing processes like casting, molding and die press forming
- Optimization of the product or process

QUESTIONS:

1. Explain the role of CAE in product design.
2. Explain the applications & benefits of CAE.
3. Enlist the capabilities of various CAE software's available in the market.

REFERENCES:

1. Computer Aided Engineering and Design by Browne Jim
2. Computer Aided Engineering Design by Saxena Anupam and Sahya Birendra

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Engineering (ME784)

Date:

EXPERIMENT NO. 2
STUDY AND APPLICATION OF COMPUTER GRAPHICS IN
CAD

AIM:

Study of computer graphics in CAD.

OBJECTIVES:

1. Review of graphics functions in C language.
2. Generation of 2D entities using C language.

THEORY:

Computer graphics can be defined as creation, storage and manipulation of pictures and drawings by means of a computer. It enhances the power of communication between the computer and its users. In a graphic mode, it is possible to display graphical figures and text. The graphic mode uses the pixel, which depends on resolution of monitor. The graphic functions are available in C language in a file 'graphics.h' which must be included in the program.

The graphics screen can be initiated by specifying the graphics driver and mode or by automatic detection. Automatic detection is the best way of initiating graphics screen and can be achieved by the following command:

```
graphdriver = DETECT  
initgraph (&graphdriver, &graphmode, " ");
```

where *graphdriver* and *graphmode* are integer data types. The graphics screen is closed by the command *closegraph* ();

When we switch over to graphics mode, cursor disappears and a coordinate system is established whereby the top left corner of the screen is treated as origin (0, 0). C++ is a powerful language for graphics and there are several routines available to create graphic image.

QUESTIONS:

1. Enlist & explain various commands/functions of graphics mode operation on your system.
2. Explain ready-made function available for drawing a point, line, arc, circle & ellipse, for setting different line styles & to write text on your system.
3. Write a computer program to draw a rectangle of sides 300 and 200 units and four circles within it, each circle touching two adjacent sides of rectangle. Assume suitable diameter of the circles.

REFERENCES:

1. Computer Aided Design by Dr. S S Khandare
2. Computer Graphics and Design by P Radhakrishnan and C P Kothandramen

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Engineering (ME784)

Date:

EXPERIMENT NO. 3
ALGORITHMS USED FOR GENERATING
2D OUTPUT PRIMITIVES

AIM:

To study routines for generating 2D output primitives.

OBJECTIVE:

Study of algorithms for displaying line and circle using C graphics/MATLAB.

THEORY:

Routines/algorithms for generating output primitives provide the basic tools for constructing pictures. Each output primitive is specified with input coordinate data and other data about the way that object is to be displayed. Points and straight-line segments are the simplest geometric components of pictures. We will study device level algorithms for displaying 2D output primitives, with particular emphasis on scan conversion methods for raster graphics system.

Line drawing is accomplished by calculating intermediate positions along the line path between two specified end point positions. An output device is then directed to fill in these positions between the end points. The process of turning on the pixels for a line segment is called 'vector generation'.

Circle is a frequently use component in pictures and graphs, a procedure for generating either full circle or circular arcs is included in most graphics packages.

QUESTIONS:

1. Draw a flow chart & prepare a computer program for scan conversion of a line using DDA algorithm.
2. Write a program to generate line with endpoints (5, 5) & (12, 15) using Bresenham's algorithm.
3. Write an algorithm to draw circle using Bresenham's method, also sketch the pixels for representing a circle with center at origin and radius = 10 units.
4. Draw a flow chart & prepare a computer program to scan convert an ellipse. Use Bresenham's method of scan conversion.

REFERENCES:

1. Computer Graphics by Donald Hearn and M Pauline Baker
2. Computer Graphics and Design by P Radhakrishnan and C P Kothandramen
3. Computer Aided Design by Dr. S S Khandare

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Engineering (ME784)

Date:

EXPERIMENT NO. 4
GEOMETRIC TRANSFORMATIONS

AIM:

To study transformation of geometric models.

OBJECTIVE:

To simulate geometric transformation of 2D entities using programming language.

THEORY:

Computer graphics image are generated from a series of line segments that are represented by the coordinates of their end points. In many applications there is need for altering or manipulating displays. Certain changes in an image can be easily made by performing mathematical operation on these coordinates, as image has been coded as numbers and stored in computer.

Geometric transformation provides a means by which an image can be enlarged in size, or reduced, rotated or moved. These changes are brought about by changing the coordinates of the image to a new set of values depending upon the requirement. Basic geometric transformations are Translation, Rotation & Scaling and other transformations include Reflection and Shearing.

QUESTIONS:

1. Write program to scale, translate, rotate & mirror the given line about origin.
2. Write program to translate and rotate triangle about origin.
3. Write a program to shear a rectangle in x & y directions.

REFERENCES:

1. Computer Aided Design by Dr. S S Khandare
2. Computer Graphics and Design by P Radhakrishan and C P Kothandramen
3. Computer Graphics by Donald Hearn and M Pauline Baker

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

EXPERIMENT NO. 5

CAE OF MANUFACTURING PROCESSES

AIM:

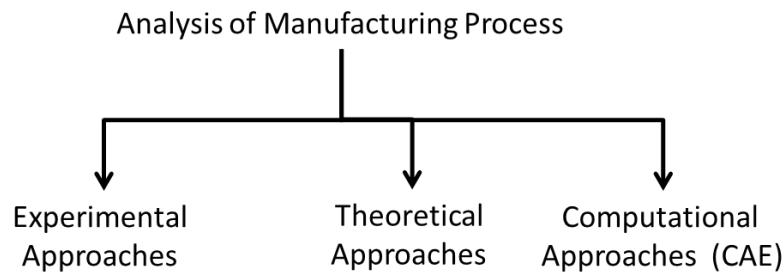
Study of different approaches used to analyze physics involved in manufacturing processes.

OBJECTIVES:

1. To study need of the CAE analysis for manufacturing processes
2. To understand physics of the manufacturing processes and identify/ formulate corresponding governing equations (algebraic, ODE, PDE)

THEORY:

There are three approaches to analyze a manufacturing process



Experimental Approach:

- Experimental approach is the oldest approach
- More accurate and reliable
- Costly and time consuming
- Measuring equipment might influence the physics of the problem

Theoretical Approach:

- This is perhaps the simplest of the approaches, but its scope is limited to simple geometries and problems
- The mathematical models governing the physics of the problem are developed to capture the behavior of the real problems
- The resulting mathematical models may be too complicated to solve analytically

Computational Approach (CAE):

- Computational approaches are mainly used to solve the governing partial differential equations of the physical process numerically
- This is mainly used to predict fluid flow, heat and mass transfer, chemical reactions, and related phenomena that occur during the manufacturing process.
- Complicated governing equations are solved numerically using a computer

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

- To address multi-disciplinary problems in industry and research, numerical approach has emerged as a powerful approach
- It is worth noting that theoretical and numerical results need to be validated to ensure its reliability

QUESTIONS:

1. List the manufacturing processes and discuss the physics involved along with their governing equations.
2. List the various CAE software programs available for analyzing manufacturing processes
3. Explain the terms pre-processor, solver and post-processor used in CAE.

REFERENCES:

1. Computational Fluid Dynamics by John D. Anderson
2. Computational Methods for Fluid Dynamics by J.H. Ferziger and M. Peric

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Engineering (ME784)

Date:

EXPERIMENT NO. 6
DISCRETIZATION METHODS

AIM:

Study of discretization methods/ schemes.

OBJECTIVES:

3. Study of spatial discretization methods
4. Study of temporal discretization methods

THEORY:

Discretization is a process of transferring continuous models and equations into discrete counterparts. This process is usually carried out as a first step toward making them suitable for numerical evaluation and implementation on computers.

	$i-1,j+1$	$i,j+1$ P	$i+1,j+1$
	$i-1,j$	i,j	$i+1,j$
	$i-1,j-1$	$i,j-1$	$i+1,j-1$

Spatial discretization of governing partial differential equation converts it in ordinary differential equation. This can be carried out by using Finite Difference Method (FDM), Finite Volume Method (FVM) or Finite Element Method (FEM).

Temporal discretization of ordinary differential equations results in system of linear algebraic equations. This system of linear equations can be solved exactly if number of unknown variables is less (typically less than 10). If number of unknown variables is large, iterative methods like **Jacobi** and **Gauss-Siedel** can be applied.

QUESTIONS:

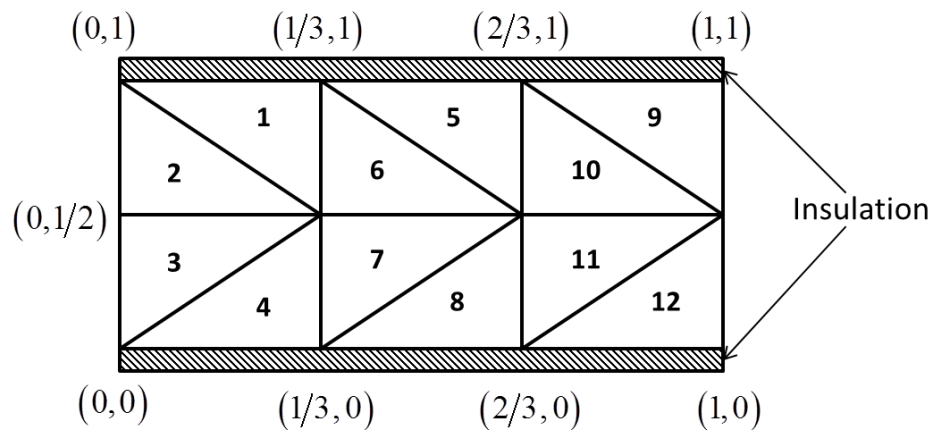
1. Explain the sources of error in numerical solution of a partial differential.
2. Discuss the advantages and limitations of spatial discretization techniques.
3. Explain the explicit and implicit approach for temporal discretization with its advantages and limitations.
4. Explain the CFL stability criteria.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

5. A square plate (size 1 x 1) of uniform thickness as shown in figure below, is divided in to 12 triangular elements. Left side of the plate is maintained at $T_L = (DD+MM)^0C$ and right side is maintained at $T_R = \sqrt{YYYY}^0C$, whereas top and bottom side of the plate are insulated thus, it involves 2D heat transfer with governing equation:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

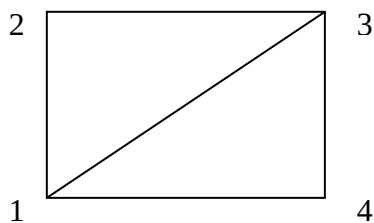
Calculate the temperature at nodes using FEM. Where DD is your Birth Date, MM is Birth Month and $YYYY$ is Birth Year.



6. Find the temperature at point P by FEM.

Point	X-coordinate	Y-Coordinate
1	0	0
2	0	1
3	1	1
4	1	0
P	$MM / 20$	$DD / 35$

Where DD is your Birth Date, MM is Birth Month and $YYYY$ is Birth Year.



$$T1 = DD, T2 = 2 * DD,$$

$$T3 = \sqrt{YYYY}, T4 = MM$$

REFERENCES:

1. Computational Fluid Dynamics by John D. Anderson
2. Computational Methods for Fluid Dynamics by J.H. Ferziger and M. Peric

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Engineering (ME784)

Date:

EXPERIMENT NO. 7
NUMERICAL SOLUTION METHODS FOR SYSTEM OF
LINEAR EQUATIONS

AIM:

Development of computer program for solution of system of linear algebraic equations.

OBJECTIVES:

1. Solution using direct methods
2. Solution using indirect/ iterative methods

THEORY:

Discretization is the process by which the closed form mathematical expressions which prescribe values at only a finite number of discrete points in the domain. Numerical methods give solution at these discrete points in the domain, called as grid points. Spatial discretization of governing partial differential equation converts it in ordinary differential equation. And implicit temporal discretization of ordinary differential equations results in system of linear algebraic equations. This system of linear equations can be solved exactly if number of unknown variables is less (typically less than 10). If number of unknown variables is large, iterative methods like **Jacobi** and **Gauss-Siedel** can be applied.

QUESTIONS:

In a numerical solution of 1-D heat conduction problem in a domain of length $L = 1\text{m}$ with end temperature of 30°C and 300°C . Considering six grid points with $\Delta x = 0.2\text{m}$, the tridiagonal matrix obtained after converting the governing differential equation to linear algebraic equation is given by:

$$\begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} T_2 \\ T_3 \\ T_4 \\ T_5 \end{bmatrix} = \begin{bmatrix} T_1 = 30 \\ 0 \\ 0 \\ T_6 = 300 \end{bmatrix}$$

By hand calculation, obtain the temperature at interior grid points (T_2, T_3, T_4, T_5) using:

- (a). Tridiagonal matrix method
- (b). Jacobi method (only 3 iterations)
- (c). Gauss-Siedel method (only 3 iterations)

Take initial guess of temperature as 100°C .

Develop computer program for all methods to compute the temperature and make sure that hand calculated results for Jacobi and Gauss-Siedel (3 iterations) matches with the computed results.

Finally, run program for a convergence criteria, $\epsilon = 10^{-3}$ and compare the number of iterations required in Jacobi and Gauss-Siedel for convergence using an initial guess of (i) 50°C , (ii) 150°C , and (iii) 200°C .

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Engineering (ME784)

Date:

EXPERIMENT NO. 8
NUMERICAL PROGRAMMING OF HEAT TRANSFER
PHENOMENA

AIM:

Development of computer program for steady heat transfer problem

OBJECTIVES:

1. Mathematical formulation of the problem
2. Solution of the governing equation using Finite Difference Method (FDM)

QUESTIONS:

1. Consider 1-dimensional steady state heat conduction in a thin sheet of thickness L . It is subjected to a boundary condition of T_1 and T_2 on the left and right face of the sheet. It is always better to do a non-dimensional analysis for a problem, Using non-dimensional parameters: $X = x / L$, and $\theta = (T - T_1) / (T_2 - T_1)$, the non-dimensional governing equation will be:

$$\frac{\partial^2 \theta}{\partial x^2} = 0$$

- (i) Write down the finite difference discretized form of the non-dimensional governing equation and non-dimensional boundary conditions.
- (ii) For a grid size, $N=5$, perform two Gauss-Seidel iterations using the calculator using an initial guess of $\theta = 0.5$.
- (iii) Develop a program to solve and obtain temperature distribution in the sheet. Compare the computed results for the grid size of 5 with the calculated results. Thereafter, calculate the temperature distribution for a grid size of 11 and 51 points. Run the code for initial guess of $\theta = 0.1, 0.5, 0.7$ and 0.9 and plot the initial guesses versus the number of iterations required for the convergence for a convergence criteria of $\epsilon = 10^{-3}$
- (iv) Results
 - a. Qualitative: Draw an overlapping plot of $\theta = f(X)$ for the various grid sizes and the exact solution
 - b. Quantitative: Calculate the non-dimensional rate of heat transfer $(qL / k(T_2 - T_1))$ for the left and right end of the sheet using 2 point one-sided numerical differentiation for the various grid sizes.
2. Repeat the above problem in the dimensional form for the thin iron ($k = 80 \text{ W/m-K}$) sheet maintained at 100°C on the left and the right end is exposed to ambient temperature of 30°C and a convective coefficient of $100 \text{ W/m}^2.\text{K}$.

Marks obtained:

Signature of faculty:

Date:

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Engineering (ME784)

Date:

EXPERIMENT NO. 09
REVERSE ENGINEERING

AIM:

Introduction to reverse engineering and its scope in CAD

OBJECTIVES:

1. To study about different types of 3-D scanners for reverse engineering.
2. To study different types of digitizing (scanning) techniques for reverse engineering.
3. To learnt how to generate a 3-D CAD model by using XOR/Redesign software.

THEORY:

Engineering is the profession of designing, manufacturing, constructing, and maintaining. In today's world of competition, we always look for the best, the most efficient, and the most durable to meet the current needs. These requirements led to the development of reverse engineering phenomenon.

Reverse Engineering, is a process that is used to create 3D CAD models directly from physical parts with little or no additional design documentation. By scanning (digitizing) items that have insufficient design documentation or require CAD data for modern manufacturing methods, a component can be quickly and accurately reverse engineered.

QUESTIONS:

1. What is reverse engineering? Justify its use.
2. Explain in brief different type of scanners? What are active scanner and passive scanner?
3. Write down specification of 3D scanner (LPX-60 model).
4. List out different reverse engineering software.
5. What are Key Benefits of Rapidform XOR/Redesign?
6. Practice a simple 3-D CAD model by using RE setup facility in the laboratory.

REFERENCES:

1. Reverse Engg. Set-up (MODROB-AICTE)
2. <http://www.rolanddga.com/asd/products/scanners/LPX60/>
3. <http://www.npd-solutions.com/reoverview.html>
4. Reversing: Secrets of Reverse Engineering by Eldad Eilamv. Publisher: Wiley
5. Rapidform XOR/Redesign tutorials.
6. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-Delhi/ComputerAidedDesign/ManufacturingII>

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Engineering (ME784)

Date:

EXPERIMENT NO. 10
RAPID PROTOTYPING

AIM:

To study about rapid prototyping.

OBJECTIVE:

1. To understand the scope and utility of rapid prototyping.
2. To study various processes of rapid prototyping.

THEORY:

Rapid Prototyping is an automatic process of manufacturing physical objects directly from their CAD models without the use of any tools, dies, molds or fixtures specific to the geometry of the objects being produced.

Rapid Prototyping improves the agility of the companies to respond to these challenges. Rapid Prototyping offers total automation in converting the virtual models into physical ones. However, use of Rapid Prototyping is still limited to the manufacture of only prototypes, mostly of non-metallic materials

QUESTIONS:

1. Explain the principle of Rapid Prototyping its advantages and applications.
2. What is soft and hard prototyping?
3. Classify and explain Rapid Prototyping processes.
4. Explain five basic system elements that affect shape of the prototype obtained in any Rapid Prototyping process.

REFERENCES:

1. Ghosh A., "Rapid Prototyping: A Brief Introduction", Affiliated East West,
2. Kenneth G. Cooper, "Rapid Prototyping Technology: Selection and Application", CRC Press, 2001.
3. Chua Chee Kai, Leong Kah Fai, Lim Chu -Sing, "Rapid Prototyping: Principles and Applications", World Scientific, 2003.

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------